Owen Thomas

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Profile

MSc Theoretical Physics student with a focus on quantum computing and information, pursuing topological quantum codes as a proposed project for my master's thesis. Proficient in Python (data analysis, hardware control and automation), quantum computation (quantum information theory, Qiskit for algorithm and circuit design) and ML (TensorFlow), with a proven ability in independent research. Interested in quantum-classical, QEC and ML based quantum circuit optimisation.

Education

King's College London, MSc Theoretical Physics — Sept 2024 - Sept 2026 (part-time)

University of Exeter, BSc (Hons) Physics — Sept 2021 – May 2024

Professional Experience

Research Intern - Experimental Optics, University of Exeter (CMRI) — June 2024 – July 2024

- Designed an effective modular architecture for construction of spectral hypercubes, configuring super-continuum lasers, tunable filters, spectrometers, cameras and flexible beam-directors on an optical workbench.
- Integrated SDKs for spectral hardware into well documented and easy to use code, enabling tailored wavelength sweeps and data collection.
- Developed an intuitive Python-based GUI to automate hardware experiments and visualise data for hyperspectral imaging research.
- Worked independently to address and overcome challenging problems and meet key milestones, verifying progress with regular meetings.

Mathematics RLHF AI Training, Outlier AI — January 2024 – June 2024

- Crafted original, graduate level mathematical prompts to challenge an LLM and force reasoning errors.
- Evaluated and refined LLM performance with high accuracy to improve mathematical reasoning via direct human feedback, taking advantage of the RLHF optimisation technique.

Academic Tutor, Self-employed — 2021 - 2024

- Provided academic tutoring to As and A-level students in Physics, Maths and Further Maths, working with each students respective strengths to deliver personalised teaching.
- Fostered a positive and encouraging environment that allowed students to explore their perceived mathematical weak-points, observing tangible improvements in their confidence and mathematical ability.

Technical Skills

Programming:

- Python (TensorFlow, Pandas, NumPy, numerical simulation, hardware control and automation)
- QisKit (quantum circuit and algorithm development)
- Machine Learning (TensorFlow for deep learning, CNN/ PINNs for classification)

Scientific Analysis:

- Experimental Physics (independent research, literature review and analysis, experiment design and implementation, data analysis, modelling and simulation)
- Theoretical Physics (quantum mechanics, quantum computation and information, QFT, general relativity, linear algebra, advanced calculus, statistics)